Course introduction

42620 Science, Technology & Society

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Part 1 teaching team from DTU Sustain

Michael
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All inquiries about part 1
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And our TAs
Karl, **Jaap**,
Seyed, **Drin**
and **Zalikha** +
invited guests.

Part 2 teaching team from DTU Management

Benjamin
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Mathieu

Anders

Tanja

Daniel

And our TAs
Emma, Alfred,
Zalikha, Jakob,
Sarah, Arantxa,
Lois, Isabela,
Emilie Frederik,
and **Amelie** +
invited guests

Why are we here?

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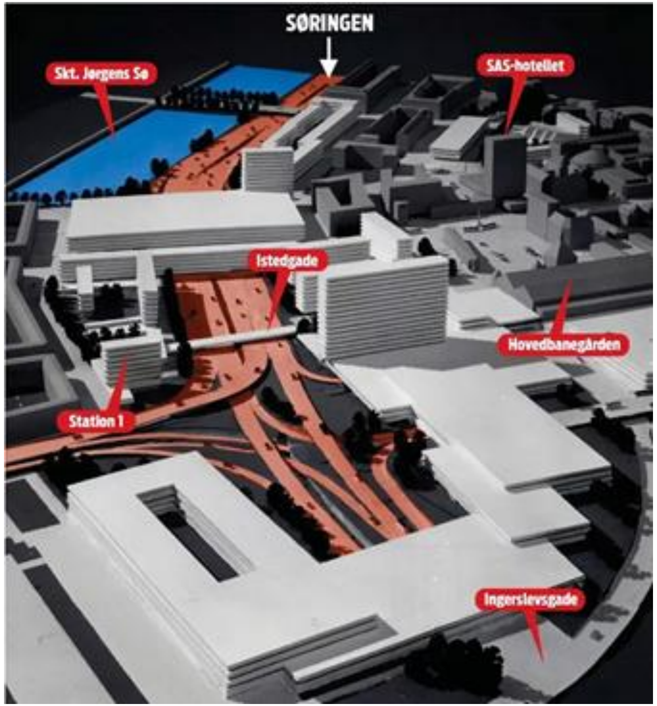
Because the history of technology suggests it might be a good idea...



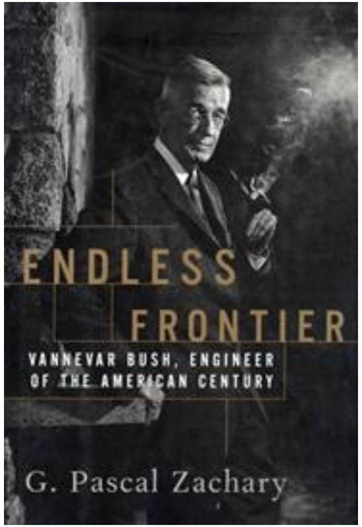
Because the Government says we have to

Because DTU needs the human-centred engineer

It is in the DNA of science to do good

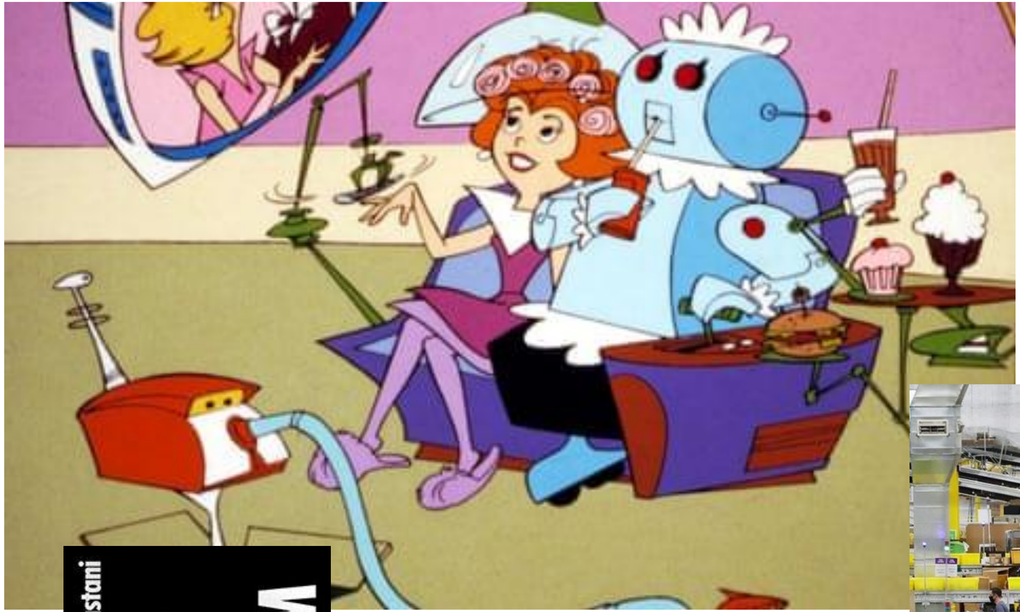
- **Francis Bacon** (17th century) founder of the scientific method, philosopher and key figure in the scientific revolution: science exists "**for the relief of man's estate**", i.e. to improve human life.
- **J.D Bernal** (1930s) physicist, inventor of X-ray crystallography, and historian of science: research has a social function; it should be **in the service of society**.
- **Vannevar Bush** (1940s) electrical engineer, inventor of the differential analyzer (early computing) and U.S. science administrator during WW2: society reaps the benefit of basic scientific research in unexpected ways and should therefore invest in it; science is an **endless frontier,** the gift that keeps on giving
- **Mariana Mazzucato** (2020s) economist and innovation adviser to the EU, science should be **mission oriented** and research focus on solving grand societal challenges, i.e. 'moonshot' projects.

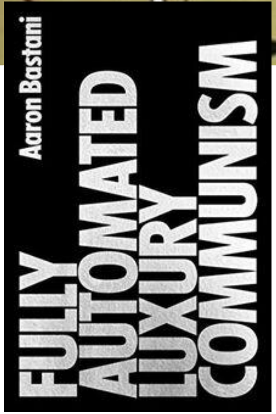
BUT: it is not always clear what is 'good' and who gets to decide it…

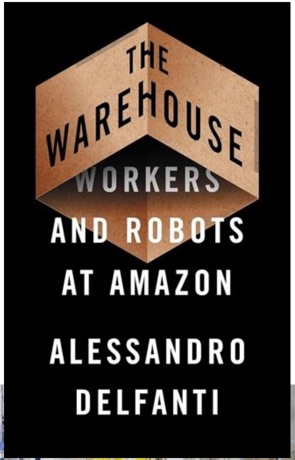



chrome


Microsoft

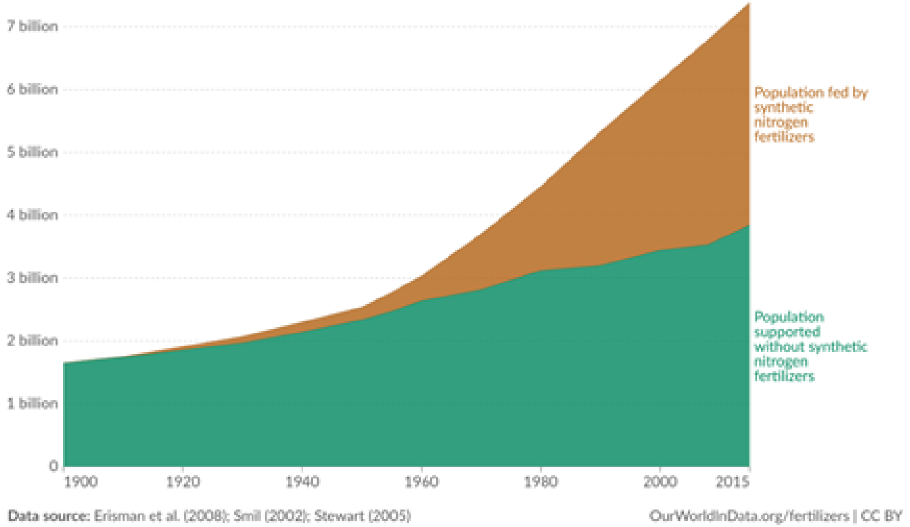



Aaron Bastani
FULLY AUTOMATED LUXURY COMMUNISM


THE WAREHOUSE
WORKERS AND ROBOTS AT AMAZON
ALESSANDRO DELFANTI

Who and what is science and technology 'good' for?

- Society needs technological solutions
- But most technological solutions also co-determine the kind of society we will have in the future
- Which makes them problems in their own right that societal actors care about



Who and what is science and technology 'good' for?

Horst Rittel and **Melvin Webber** (1973), design theorists, urban planners, and founders of the theory of **wicked problems**:

"In a pluralistic society there is nothing like the undisputable public good; there is no objective definition of equity; policies that respond to social problems cannot be meaningfully correct or false; and ***it makes no sense to talk about "optimal solutions" to social problems unless severe qualifications are imposed first.****"*

Horst Rittel ▲

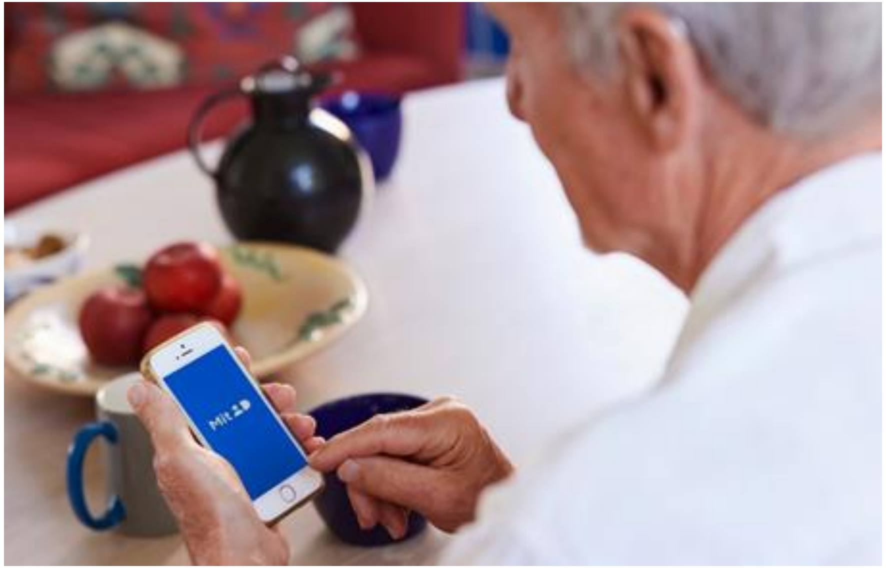
▼ *Early operations research (WWII)*

On this course you will first **(part 1)** learn to assess what is objectively more sustainable solutions and then **(part 2)** learn to responsibly deal with the reality that different actors in society have different ideas than you about what counts as a good solution.

The problems where science and technology are supposed to help us are very often “wicked”:

- Unintended and unforeseen societal consequences (positive and negative)
- Disagreement on problem definitions
- Uncertain knowledge and contested expertise
- Attempts to delimit the problem is effectively to take sides in the discussion
- Attempts to close a discussion by reference to the facts will be met with scepticism
- Existing in a polarized, chaotic and sometimes manipulative information environment.


In your dreams
Reality…



As an engineer, you need to navigate such situations

- To responsibly implement solutions.
- To design technology for people with different stakes and values.
- And to be able to contribute your expertise when it is needed and to those who need it.

"An economic disaster": New report slams against Energy Island Bornholm

A new report from the think tank Kraka Economics takes a critical stance on the potential realization of Energy Island Bornholm, which could be an economic disaster

Energy island – the Baltic sea's nodal point for intelligent energy

CIP fonden

The CIP Foundation shows the way to an interconnected Danish hydrogen infrastructure

September 11, 2023

WINDPOWER MONTHLY

Official documents reveal vulnerability of Danish energy island to sabotage

The vulnerability of Denmark's proposed offshore energy infrastructure to sabotage has been revealed in documents produced by its climate and energy ministry, the Danish Energy Agency (DEA) and grid operator Energinet.

The lack of offshore wind bids in the North Sea has shaken the government and Denmark's green transition

12. December 2024

Case in point: The Danish energy islands

A course in two parts

PART 1: What is objectively the most sustainable way to build the energy islands?

You learn to assess the sustainability of technological projects using life cycle assessment and SDG methodology.

This is an **expert-driven approach** where we have already decided what is in society's best interest.

Output by the end of the week: Your suggestions for how to improve the sustainability of the energy islands.

PART 2: What will real world stakeholders with other priorities than us think about our proposals for improved sustainability? And how can we include their concerns?

You learn to map societal debates about technological projects, analyse the arguments and recommend strategies responsible innovation that takes those arguments seriously.

This is a **bottom-up, empirical approach** where we let other actors define what is in society's best interest.

Output by the end of the course: A policy brief on how to responsibly implement your proposals for improved sustainability considering existing disagreement between actors.

EXAM (1/2): Multiple-Choice Test

This is an **on-site** individual exam in the morning of the last day of the course (23/1). Evaluated individually as passed or failed.

Consists of questions regarding the concepts learnt in the lectures of the 2nd week.

MCT

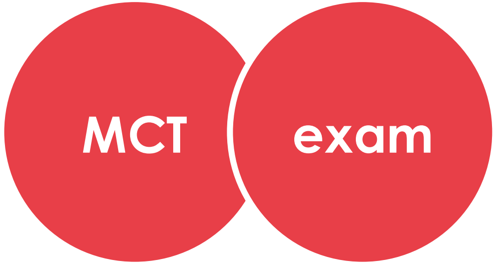
EXAM (2/2): Write a policy brief for a real world stakeholder

Submitted in groups on the last day of the course (23/1) with individual attribution of responsibility.

Evaluated individually as passed or failed.

Three days (20/1-23/1) set aside for writing and supervision (collective Q&A).

Each group will be assigned a supervisor who will also grade their exam.

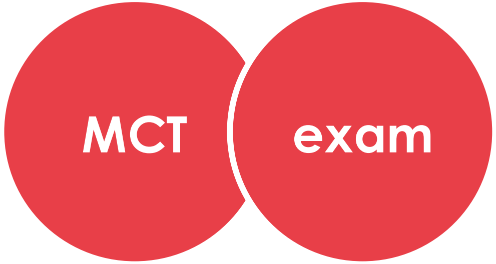


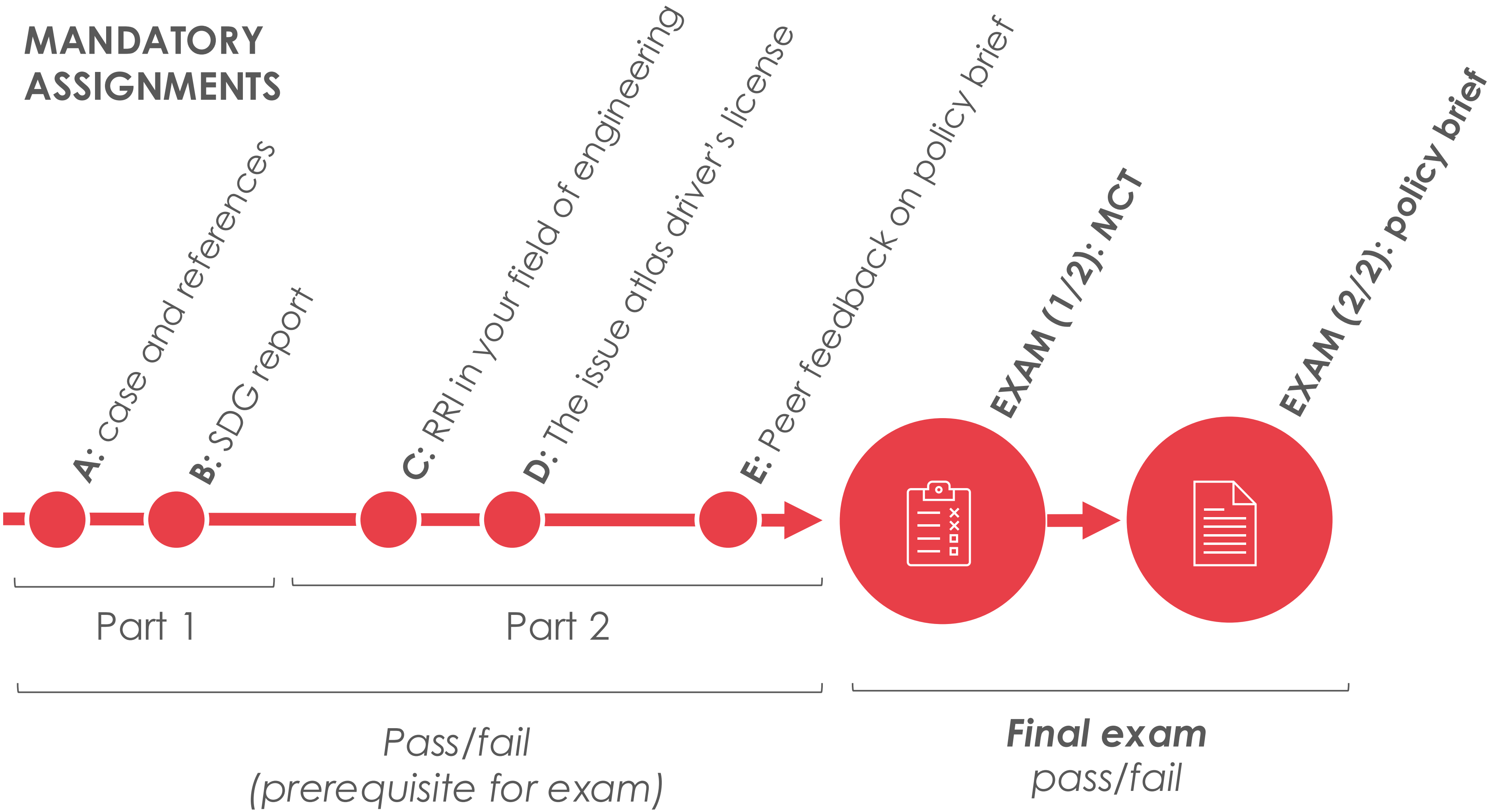
EXAM (2/2): Write a policy brief for a real world stakeholder

Suggest ways to improve the sustainability of the energy islands (taken from Part 1).

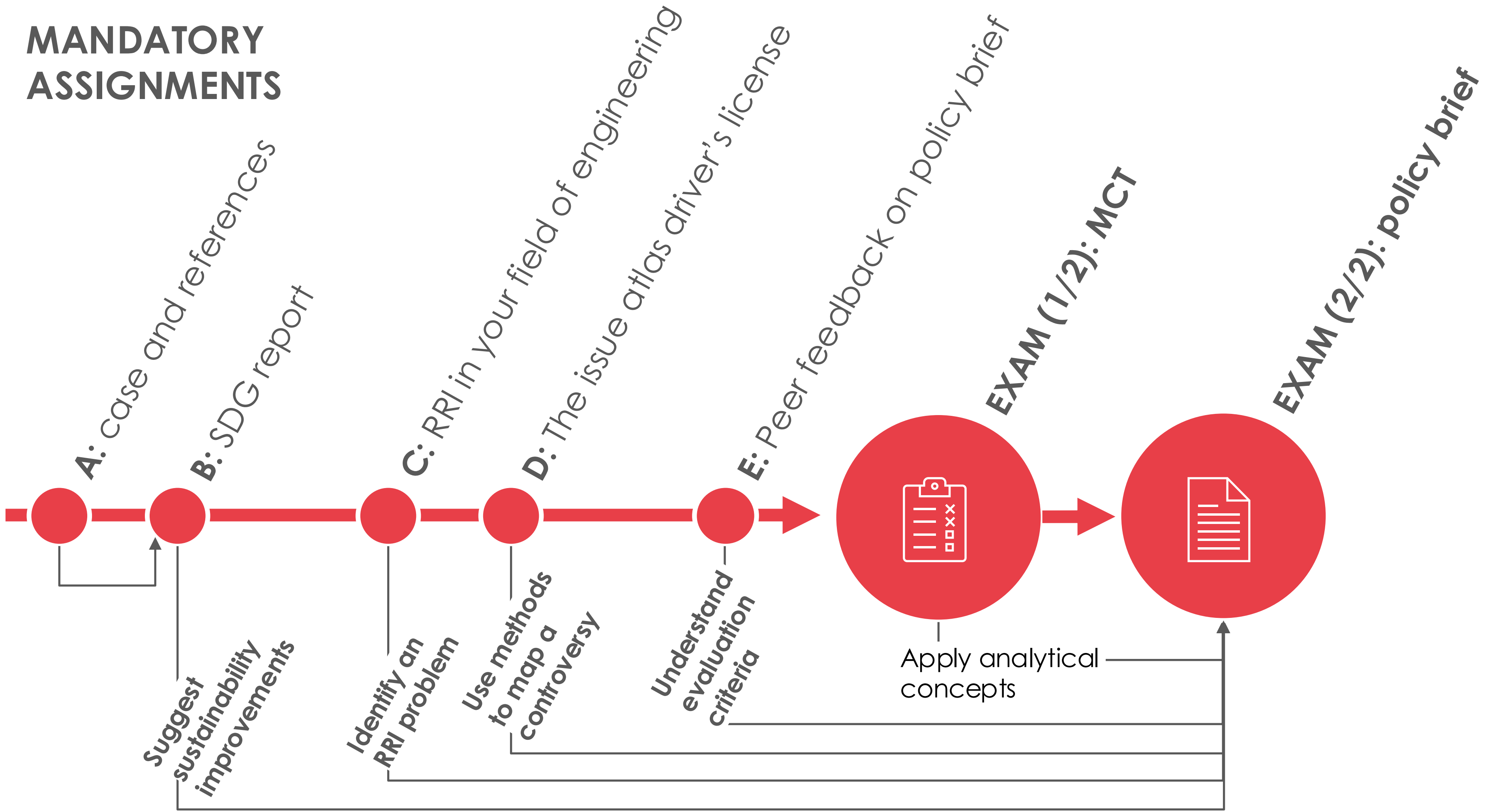
Map and analyse issues and actor positions in the energy island controversy and assess their implications for responsible innovation in relation to your suggested sustainability improvements.

Recommend a strategy for responsible implementation/development of your sustainability improvements.




MANDATORY ASSIGNMENTS
A: case and references
B: SDG report
C: RRI in your field of engineering
D: The issue atlas driver's license
E: Peer feedback on policy brief
EXAM (1/2): MCT
EXAM (2/2): policy brief
Part 1
Part 2
Pass/fail
(prerequisite for exam)
Final exam
pass/fail

MANDATORY ASSIGNMENTS



Learning objectives

- **Establish the significance of SDGs in relation to engineering projects and discover how they can be used in practice and analysis.**
- **Know and understand prevailing definitions of sustainability and the history of the concept of sustainability.**
- **Understand the importance of adopting a life cycle/systems perspective and considering a comprehensive range of impacts within each of the three sustainability dimensions - environment, society and economy.**
- Identify how social values are embedded in engineering projects and how technology can become a matter of public concern.
- Describe the role of ethics in engineering work within one's own engineering field and in collaboration with other fields and stakeholders.
- Describe and evaluate the primary social processes of designing, implementing, adopting, and contesting innovation.
- Discuss key dimensions of responsible research and innovation and apply them to evaluate engineering projects, especially anticipation, reflexivity, inclusion, and responsiveness.
- Recommend responsible strategies to foster absolute social and environmental sustainability of engineering projects.

PART 1
Assignments **A** and **B**

Learning objectives

- Establish the significance of SDGs in relation to engineering projects and discover how they can be used in practice and analysis.
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- **Discuss key dimensions of responsible research and innovation and apply them to evaluate engineering projects, especially anticipation, reflexivity, inclusion, and responsiveness.**
- **Recommend responsible strategies to foster absolute social and environmental sustainability of engineering projects.**

PART 2
Assignments **C, D,** and **E**

Mandatory assignments

- All mandatory assignments must be passed to be able to partake in the final exam assignments.
- Assignments are completed in groups, except the MCT which is individual.
- Evaluated by the teaching team immediately after you hand in.
- Evaluation is pass/fail.
- Feedback within a couple of days.
- If you fail an assignment, the feedback will tell you what to improve.
- Revised assignments receive a new evaluation until passed (within course timeframe and provided deadlines)
- Failure to complete revisions = failed assignment = no exam.

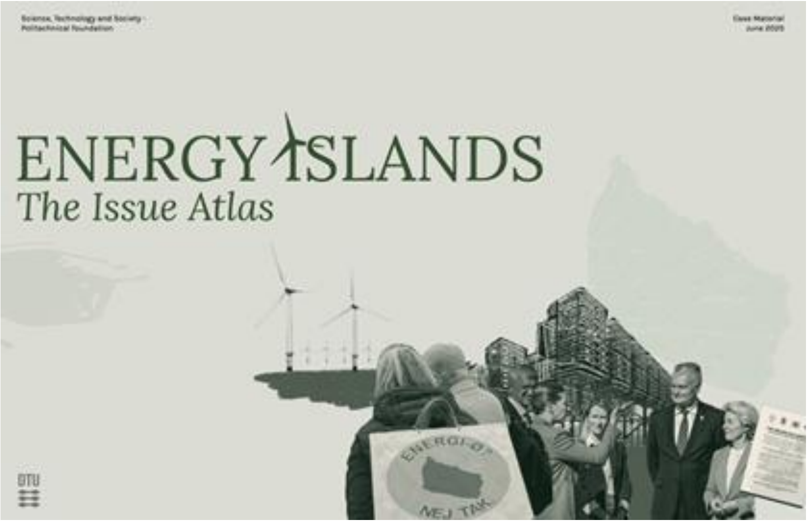
Readings

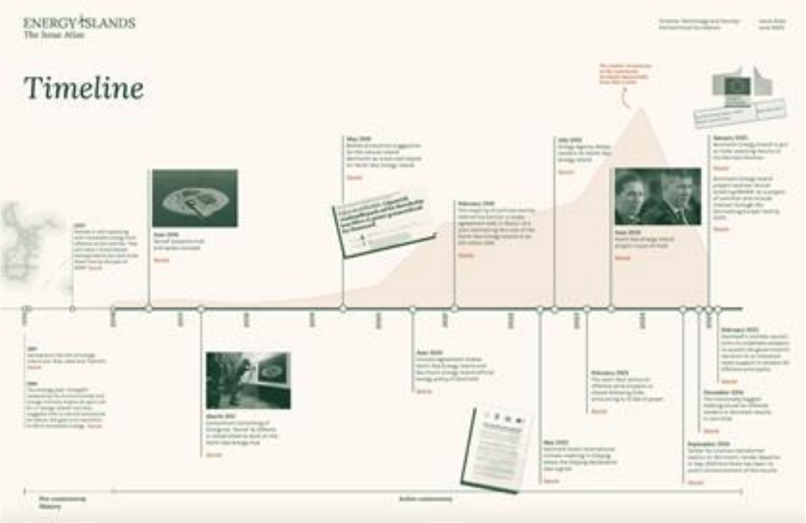
- No readings and nothing to buy in advance of the course.
- All materials will be available on Learn or in links to open sources.
- For Part 1: Two key readings to be consulted during the week (already available on Learn).
- For Part 2: Lecture notes summarizing main points (need to know) with optional suggested readings (nice to know).

The Issue Atlas

Case material designed for this course.

- A **report** detailing the political, social and technological context of the case, a description of key actors and their roles (all of which will be available from the beginning of the course), an overview of methods used to generate the dataset and a set of baseline visualizations (the latter will be available in part 2 of the course)
- A set of **key documents** in the case (available from the beginning of the course)
- A **dataset** of statements made by different actors in the public controversy over the islands (available in part 2 of the course)
- A set of **python scripts** for exploring the dataset and generate visualizations (available in part 2 of the course)





Other practical course info

- TAs are there to help. Please use the dedicated queuing system to book them (link will be available on Learn).
- Assignments, slides, and other lecture materials will become available on Learn as they approach. Stay updated.
- **NB:** This is a course under constant development and you cannot count on using course materials from previous iterations. Responding to student feedback we have changed several of the mandatory assignments, the final assignment, and reordered much of the lecture content in Part 2.
- *Please provide feedback directly to us during the course if information is missing, clarification is needed, or improvements can be made on the go.*

Good luck!